

# Guided Inquiry: Exploring the Reynolds Number

**Objective:** Use the interactive Reynolds Number Explorer to understand how fluid properties, velocity, and pipe diameter determine flow regimes.

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## Part 1: Understanding Viscosity (Dynamic vs. Kinematic)

Before calculating flow, we must distinguish between the two types of viscosity displayed in the tool.

### 1. Dynamic Viscosity ( $\mu$ , **mu**):

- **Definition:** This is the fluid's internal resistance to flow (thickness). It measures how much force is needed to slide layers of fluid past each other.
- **Unit:** Poise (P) or g/(cm·s).
- *Analogy:* How hard it is to stir a spoon through the fluid.

### 2. Kinematic Viscosity ( $\nu$ , **nu**):

- **Definition:** This is the ratio of dynamic viscosity to density ( $\nu = \mu/\rho$ ). It measures how easily momentum diffuses through the fluid.
- **Unit:** Stokes (St) or cm<sup>2</sup>/s.
- *Key Concept:* In the Reynolds equation ( $Re = \frac{V \cdot D}{\nu}$ ), kinematic viscosity is often the "governing" factor for how likely a fluid is to become turbulent.

### Activity:

Select **Water** and then **Honey** in the simulator. Record their values below:

| Fluid | Density ( $\rho$ ) | Dynamic Viscosity ( $\mu$ ) | Kinematic Viscosity ( $\nu = \mu/\rho$ ) |
|-------|--------------------|-----------------------------|------------------------------------------|
| Water |                    |                             |                                          |
| Honey |                    |                             |                                          |

- **Question 1:** Honey is much denser than water, but its dynamic viscosity is massive. How does this affect its Kinematic Viscosity compared to water?
- **Question 2:** Based on the formula  $Re = \frac{V \cdot D}{\nu}$ , does a *higher* kinematic viscosity make it easier or harder for flow to become turbulent?

## Part 2: Fluid Comparison Experiment

**Setup:** Set the **Pipe Diameter (D)** to **5.0 cm** and **Velocity (V)** to **100 cm/s** for all tests below.

### Experiment A: Water

1. Select **Water**.
2. Record the Reynolds Number ( $Re$ ): \_\_\_\_\_
3. Is the flow Laminar, Transitional, or Turbulent? \_\_\_\_\_

### Experiment B: Honey

1. Select **Honey** (keep  $D = 5.0$ ,  $V = 100$ ).
2. Record the Reynolds Number ( $Re$ ): \_\_\_\_\_
3. **Observation:** Try to increase the Velocity to the maximum (500 cm/s). Can you get Honey to become turbulent? Why or why not?

### Experiment C: Air

*Note: Many students assume air is "thinner" and thus always more turbulent. Let's test this.*

1. Select **Air** (Reset to  $D = 5.0$ ,  $V = 100$ ).
2. Record the Reynolds Number ( $Re$ ): \_\_\_\_\_
3. **Analysis:** Look at the Kinematic Viscosity of Air compared to Water. Which is higher? Explain why Air has a lower Reynolds number than Water under identical conditions.  
*Hint: Remember that Air has very low density (inertia), so it struggles to overcome its own viscosity to create turbulence compared to water.*

## Part 3: The Shape of Flow (Laminar vs. Turbulent)

**Setup:**

- Select **Engine Oil**.
- Set **Diameter** to **4.0 cm**.
- Adjust **Velocity** until  $Re$  is approx **1000** (Laminar).

## Observation 1: The “No-Slip” Condition

Look closely at the blue flow lines moving through the pipe.

1. Compare the speed of the lines in the **center** of the pipe vs. the lines near the **top and bottom walls**.
2. Where is the fluid moving fastest? \_\_\_\_\_
3. Where is the fluid moving slowest? \_\_\_\_\_

### Theory:

This is called the **Parabolic Velocity Profile**. In real physics, fluid molecules touching the wall have a velocity of zero (they are stuck). This friction slows down the layers near the wall, while the center flows freely.

## Observation 2: Transition to Chaos

1. Slowly increase the **Velocity**.
2. Watch what happens when  $Re$  crosses **2000**.
3. Describe the change in the particle lines. Do they still follow straight paths?
4. Increase  $Re$  above **4000**. What happens to the “layers”? Do you still see a clear difference between the wall speed and center speed, or is it all mixed up?

## Conclusion

Summarize the relationship between the variables:

- Increasing **Velocity** \_\_\_\_\_ the Reynolds Number.
- Increasing **Diameter** \_\_\_\_\_ the Reynolds Number.
- Increasing **Viscosity** \_\_\_\_\_ the Reynolds Number.